Evidence-Gated Stabilization for Sparse-View 3D Gaussian Splatting

Tanveer Younis[a,b], Zhanglin Cheng[a,*]

***Abstract*—Sparse-view 3D Gaussian Splatting can fit input images with primitives that appear plausible near the training trajectory but become floaters, billboard-like structures, or drifting geometries under larger viewpoint changes. We view this behavior as an optimization-dynamics problem: standard 3DGS applies the same geometry updates and densification rules to primitives that are well supported by multiple views and to those that are under-constrained by a low parallax. We propose *Evidence-Gated Stabilization*, which uses primitive-level multiview support to regulate sparse-view 3DGS training. From a standard sparse-view initialization, we estimated an *Angular Support Score* using scaffold-consistent visibility, pairwise parallax, and baseline-to-depth normalization. This score modulates the updates of the Gaussian position, scale, and rotation, restricts splitting and cloning, and guides pruning and opacity refinement, while leaving the appearance parameters adaptive. Our method reduces the influence of primitives that lack sufficient support. Experiments on Tanks and Temples, DL3DV-10K, and Mip-NeRF 360 showed improved PSNR, SSIM, and LPIPS over sparse-view 3DGS baselines, and qualitative comparisons indicated fewer off-trajectory floaters and billboard-like artifacts.**

***Index Terms*—3D Gaussian Splatting, sparse-view reconstruction, novel view synthesis, geometric consistency, support-gated optimization**

I. Introduction

3D Gaussian Splatting (3DGS) has become a widely used representation for real-time novel view synthesis, combining explicit Gaussian primitives with efficient differentiable rasterization [1]. Under dense camera coverage, photometric supervision typically provides sufficient redundancy to reliably optimize the geometry, opacity, and appearance. However, this assumption breaks down in sparse-view settings: a small number of input images may be explained by primitives that fit the training views but are poorly constrained as 3D structures.

This instability can be underestimated when the evaluation emphasizes views close to the training trajectory. When the test views remain near the training cameras, view-dependent billboards and weakly constrained floaters may still project plausibly, producing acceptable image-space quality. The failure becomes more apparent when the camera moves away from the input view manifold. Under such off-trajectory viewpoints, weakly supported primitives may become visible as floating debris, planar stretching, background collapse, or geometric drift. We therefore use off-trajectory rendering as a stress test to determine whether a sparse-view 3DGS representation remains coherent beyond local view interpolation.

Prior sparse-view 3DGS methods typically address this problem by adding external supervision, regularization, or post hoc cleanup. Depth-regularized approaches introduce monocular or multi-view depth priors [2], [3], [4]; geometry-aware methods incorporate epipolar, covisibility, or matching constraints [5], [6], [7]; and pruning- or density-control strategies reduce redundant or unreliable Gaussians during or after optimization [8], [9]. These methods improve sparse-view reconstruction; however, in many pipelines, the optimizer and topology update rules still treat primitive geometry updates as if they were equally reliable across well-supported and ambiguous regions. As a result, a Gaussian may move, stretch, split, accumulate opacity, and persist even when its geometry is supported only by narrow-baseline or inconsistent observations.

We argue that sparse-view 3DGS failure should therefore be viewed not only as a regularization problem but also as an evidence-allocation problem. In dense-view settings, most primitives receive enough multi-view support that photometric gradients are often a reasonable proxy for geometric improvement. In sparse-view settings, this assumption breaks down. Some primitives are observed with sufficient angular diversity, while others are under-constrained along the viewing rays. Nevertheless, standard optimization applies the same geometric degrees of freedom to both. The problem is amplified by 3DGS topology growth: a weakly supported primitive may be moved by ambiguous photometric gradients and then cloned or split, turning local ambiguity into a larger set of visible floaters or billboard-like structures.

Our key idea is to use multiview geometric evidence as an explicit control signal for sparse-view 3DGS optimization. Instead of optimizing primitives solely according to photometric error, we also estimate whether the sparse input views provide enough evidence to support their geometric updates and topology changes. In this formulation, evidence regulates three aspects of representation. It modulates how much a primitive moves by scaling geometry gradients, regulates topology growth by restricting splitting and cloning to sufficiently supported regions, and guides persistence through pruning and opacity refinement for weakly supported Gaussians. This principle augments photometric fitting with evidence-aware control over geometry and topology.

To operationalize this principle, we introduce *Evidence-Gated Stabilization*. In our evaluation pipeline, MASt3R [10] provides an initial geometric scaffold and camera estimates, which are registered within a common coordinate frame before

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Fig. 1. Near-trajectory vs. off-trajectory novel views under sparse inputs. Top: Near-trajectory views close to the training camera manifold, where FSGS [4], SparseGS [3], NexusGS [7], and our method produce visually plausible renderings. Bottom: Off-trajectory views with larger viewpoint deviation, where baseline methods exhibit billboard-like distortions and floating artifacts. Our Evidence-Gated Stabilization uses multi-view geometric support to reduce the influence of weakly supported primitives, yielding fewer visible off-trajectory artifacts.

initializing the Gaussian representation. Rather than treating this scaffold as the final reconstruction, we use it as an evidence substrate. For each Gaussian primitive, we compute an *Angular Support Score* that integrates scaffold-consistent visibility, pairwise angular parallax, and baseline-to-depth normalization. This score quantifies the relative multiview support available to each primitive under the given scaffold and camera configuration.

We then use this support signal to regulate both optimization and topology. In Support-Gated Optimization, the support score modulates the gradients of the geometric parameters, including position, scale, and rotation, while leaving the appearance parameters free to adapt to local radiance and view-dependent effects. This decoupling reflects the observation that sparse-view ambiguity is especially harmful for geometric placement, while appearance parameters can still benefit from photometric adaptation. In Evidence-Guided Topology Adaptation, the same support signal restricts splitting, cloning, pruning, and support-aware opacity refinement. As a result, representational capacity is biased toward regions with stronger multi-view evidence, while weakly supported primitives are discouraged from expanding into persistent artifacts. This design assumes that the initialization and camera poses provide usable candidate geometry. If the scaffold omits or corrupts a structure, the resulting support estimate may inherit that error and suppress its recovery.

In summary, our contributions are:

- We analyze sparse-view 3DGS instability as an optimization-dynamics problem in which weakly supported primitives can move, grow, and persist under ambiguous photometric gradients.
- We introduce a primitive-level Angular Support Score that estimates multi-view geometric support from scaffold consistency, angular parallax, and baseline-to-depth normalization.
- We propose support-gated geometry optimization and evidence-guided topology adaptation to regulate primitive motion, growth, and persistence while keeping appearance optimization adaptive.

II. Related Work

Sparse-view neural rendering. NeRF and related implicit representations achieve high-quality novel view synthesis when views are sufficiently dense [11], [12], [13]. With sparse inputs, photometric supervision becomes under-constrained, and prior work mitigates this ambiguity with unseen-view regularization, semantic or multi-view priors, and scene parameterizations for unbounded trajectories [14], [15], [16], [17]. 3DGS [1] explicitly models scenes using optimizable anisotropic Gaussian primitives, but its flexibility can amplify sparse-view ambiguity: geometry and topology can change to fit the input views while remaining poorly constrained in 3D structure. Our work focuses on this optimization-dynamics aspect of sparse-view 3DGS.

Sparse-view 3D Gaussian Splatting. Recent sparse-view 3DGS methods improve stability by adding external priors, regularization terms, or stronger initialization. Depth-based approaches introduce monocular or multi-view depth constraints to reduce geometric ambiguity [18], [4], [2], [3]. Matching- and geometry-aware methods incorporate cross-view correspondences, covisibility maps, epipolar depth priors, or co-regularization to improve consistency across sparse inputs [19], [20], [6], [5], [7], [21]. Initialization-oriented pipelines use learned dense matching or reconstruction priors to obtain stronger sparse-view starting points [22], [23], [24], [25]. Stochastic or regularization-based variants, such as dropout-style training, further reduce overfitting to the input views [26]. These methods are complementary to our goal. Instead of proposing another external prior or initializer, we use multi-view support as a primitive-level reliability signal during optimization. The key distinction is that the signal directly regulates how each Gaussian is allowed to move, grow, and persist during training.

Density control, pruning, and reliability. The quality and size of a 3DGS model depend on densification, splitting, pruning, and opacity updates. Prior work improves primitive models, detail recovery, pixel-aware densification, compact representations, and safe pruning [27], [28], [29], [30], [8], [9]. In sparse-view settings, however, a large image-space gradient may indicate either missing detail or geometric conflict. We

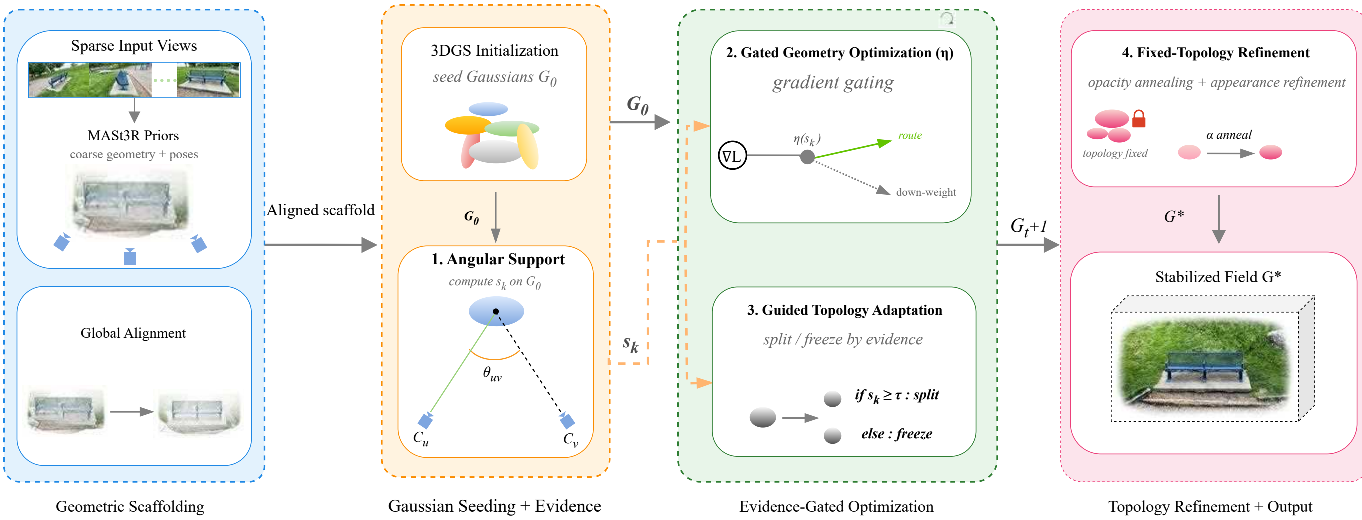

Fig. 2. Overview of the proposed pipeline. Starting from a standard sparse-view initialization, we use coarse scaffold depths and input cameras to compute a primitive-level Angular Support Score s_k. This score controls three training-time operations: geometry update gating, support-aware topology adaptation, and fixed topology opacity refinement. The scaffold is used as an evidence substrate; the contribution is how multi-view support regulates primitive motion, growth, and persistence during optimization.

therefore gate topology growth with angular support, so capacity is preferentially allocated to primitives with stronger multi-view evidence. This idea is related to classical multi-view reliability cues: Viewing angle, baseline, depth, visibility, and view selection are established multi-view reliability cues [31], [32], [33]. Our contribution is to use such cues as a per-Gaussian training signal that modulates geometry updates, densification, pruning, and opacity refinement.

Geometric initialization and feed-forward methods. Classical SfM/MVS systems estimate camera geometry and scene structure from multi-view observations [34], [33]. More recent sparse-view 3DGS pipelines employ learned reconstruction priors, such as DuSt3R [35] and MASt3R [10], to obtain camera estimates and dense geometric initialization; InstantSplat is a representative example [23]. Our evaluated pipeline specifically uses a MASt3R-derived scaffold and camera estimates. Unlike initialization-oriented methods, our contribution operates after initialization by using per-Gaussian multiview support to regulate geometry gradients and topology throughout training. Feed-forward methods predict views or Gaussian representations directly from sparse context images [36], [37], [38], [39], [40], [41], [42]. They target cross-scene generalization and fast inference, whereas our setting is per-scene optimization from a fixed sparse capture. Our method is intended to improve the geometric stability of that optimization process.

III. Method

A. Overview

Sparse-view 3DGS is unstable because the photometric gradients are not equally reliable for all Gaussian primitives. In regions with sufficient multiview parallax, photometric updates can provide useful geometric guidance. However, in weakly observed or low-parallax regions, the same updates may move, flatten, densify, or preserve primitives in ways that reduce training-view error but yield unstable geometry under novel viewpoints. The standard 3DGS applies the same geometry update and topology growth rules in both cases.

We propose *Evidence-Gated Stabilization*, a training-time framework that uses primitive-level multiview support as an explicit control signal. In our evaluated pipeline, MASt3R [10] provides the camera parameters and a coarse per-view geometric scaffold. Following global alignment, this scaffold is used to initialize the Gaussian set G_0. We use the resulting geometry to estimate the support of each primitive and regulate its motion, growth, and persistence during optimization. Our experiments establish the method only under this MASt3R-derived initialization; compatibility with other scaffold providers is not evaluated. In our evaluated pipeline, MASt3R produces the per-view scaffold geometry and camera estimates. The "Global Alignment" block in Fig. 2 registers these estimates into a common coordinate frame before initializing G_0.

As illustrated in Fig. 2, the method has four main components. First, we compute a primitive-level Angular Support Score s_k from scaffold-consistent visibility, pairwise angular parallax, and baseline-to-depth normalization. Second, we used this score to gate geometry updates for the Gaussian position, scale, and rotation, while leaving the appearance parameters adaptive. Third, we used the same support signal to guide topology adaptation by restricting splitting and cloning in weakly supported regions and pruning low-opacity primitives with persistently low support. Finally, during fixed-topology refinement, we attenuate the opacity of the primitives that remain weakly supported over time. The support score is a soft reliability estimate and not a proof of physical correctness.

B. Primitive-Level Angular Support

We estimate how strongly each Gaussian primitive is supported by the sparse input views. The score is computed from the current Gaussian center, input cameras, and coarse scaffold depth. It is intended to indicate whether a primitive has enough multi-view evidence to justify geometry updates and topology growth.

Scaffold-Consistent Observation. Let G_k be the k-th Gaussian with center μ_k. For a training view v, let $\pi_v(\cdot)$ denote perspective projection, Ω_v the valid image region, $z_{k,v}$ the depth of μ_k in the camera frame of v, and $\tilde{z}_v(\cdot)$ the scaffold depth map sampled at the projected pixel. We define

$$\mathbb{V}_{geo}(k,v) = \mathbf{1}[\pi_v(\mu_k) \in \Omega_v] \cdot \mathbf{1}[|z_{k,v} - \tilde{z}_v(\pi_v(\mu_k))| \leq \delta_z], \tag{1}$$

where δ_z is a depth tolerance defined relative to the scene scale. The first indicator verifies that the Gaussian center projects inside the valid image region, while the second checks whether its camera-space depth agrees with the scaffold depth at the projected pixel. A view contributes angular evidence only when both conditions are satisfied. This is a coarse consistency test rather than a complete visibility model; it does not explicitly handle transparency, multi-layer geometry, fine occlusion boundaries, or scaffold errors. The set of scaffold-consistent observing views is

$$\mathcal{V}_k = \{v \mid \mathbb{V}_{\text{geo}}(k,v) = 1\}. \tag{2}$$

If $|\mathcal{V}_k| < 2$, the primitive has insufficient multi-view angular support.

Angular Support Score. Mere visibility is insufficient for robust optimization; the observing views must also provide useful parallax. Inspired by multi-view triangulation reliability [32], [33], we combine an angular term with a baseline-to-depth normalization. For each primitive, the Angular Support Score s_k is formulated as the normalized average pairwise support:

$$s_k = \begin{cases} 0, & \text{if } |\mathcal{V}_k| < 2, \\ \dfrac{1}{|\mathcal{V}_k|(|\mathcal{V}_k|-1)} \displaystyle\sum_{\substack{u,v \in \mathcal{V}_k \\ u \neq v}} \underbrace{\frac{||C_u - C_v||_2}{||\mu_k - C_u||_2 + ||\mu_k - C_v||_2}}_{\text{Resolution Confidence}} \cdot \underbrace{\sin(\theta_{uv}^{(k)})}_{\text{Triangulation}}, & \text{otherwise.} \end{cases} \tag{3}$$

where C_u and C_v denote the camera centers, and $\theta_{uv}^{(k)}$ is the angle between their viewing rays toward μ_k. The baseline-to-depth factor measures camera separation relative to the primitive's distance, assigning lower support to distant primitives observed from a narrow baseline. The $\sin(\theta_{uv}^{(k)})$ term penalizes nearly collinear rays, which provide weak triangulation constraints. Averaging over all valid view pairs prevents the score from increasing solely with the number of observing views. Scores are normalized over active primitives at each support refresh before being used for gating and opacity refinement.

Adaptive Thresholding. Because parallax distributions vary across scenes and view layouts, we compute support thresholds from the empirical score distribution: $\tau_{low} = Q_{20}(\{s_k\})$ and $\tau_{mid} = Q_{50}(\{s_k\})$. These quantiles adapt the gate to scene-dependent support distributions, but they should be interpreted as relative reliability estimates rather than absolute geometric guarantees. The normalized support used for gating is

$$\eta_k = \text{clamp}\left(\frac{s_k - \tau_{low}}{\tau_{mid} - \tau_{low}}, 0, 1\right). \tag{4}$$

C. Support-Gated Geometry Optimization

Standard 3DGS updates all primitives according to photometric residuals. In sparse settings, this can move weakly supported primitives in geometrically ambiguous directions. We therefore use η_k as a geometry gate. Low-support primitives receive suppressed photometric geometry gradients, while high-support primitives receive full geometry updates.

We apply this gate only to geometric parameters $\Theta_{geo} = \{\mu, S, q\}$ (S and q denote scale and rotation). Appearance parameters, such as spherical harmonic coefficients, are optimized normally. This decoupling reflects that sparse-view ambiguity is especially harmful for geometric placement, while appearance can still benefit from photometric adaptation.

Let $\mathcal{L}_{photo}$ be the standard 3DGS photometric loss and $\mathcal{L}_{reg}$ the geometry regularization term. We update geometry using

$$\Theta_{geo}^{(t+1)} \leftarrow \Theta_{geo}^{(t)} - \lambda \mathcal{U}\big(\eta_k \nabla_{\Theta_{geo}} \mathcal{L}_{photo} + \beta \nabla_{\Theta_{geo}} \mathcal{L}_{reg}\big), \tag{5}$$

where t denotes the iteration step, λ is the learning rate, β is the regularization weight, ∇ is the gradient operator, and $\mathcal{U}(\cdot)$ denotes the optimizer's per-parameter update transform (Adam in our implementation, matching the official 3DGS codebase). Because the regularization term is not multiplied by η_k, $\eta_k = 0$ should be understood as suppressing the current photometric geometry gradient, not as freezing all parameter changes. In our implementation, the gate is applied to raw geometry gradients before the Adam update. Every $T_{\text{supp}} = 500$ iterations, we recompute each primitive's support score from its current center, the fixed input cameras, and the scaffold depths; we then renormalize the scores and update the percentile thresholds over the active Gaussian population. The resulting values are cached between recomputations, and newly split or cloned primitives inherit their parent's support value until the next recomputation.

Anisotropy Regularization. Anisotropic Gaussians are useful for representing surfaces, so we do not enforce isotropy. Instead, we penalize only excessive aspect ratios that can contribute to flattened or elongated artifacts:

$$\mathcal{L}_{reg} = \mathcal{L}_{ani} = \sum_k \max\left(0, \frac{\max(S_k)}{\min(S_k)} - \kappa\right), \tag{6}$$

where $S_k \in \mathbb{R}^3$ contains the scale components of G_k, and κ is a threshold ratio.

D. Evidence-Guided Topology Adaptation

In standard 3DGS, densification is driven primarily by view-space positional gradient magnitude. Under dense supervision, high gradients often indicate missing detail. Under sparse supervision, however, high gradients can also arise from geometric conflict or low-parallax ambiguity. We therefore use support-aware topology adaptation to regulate primitive growth and persistence.

Support-Aware Densification. A primitive G_k is eligible for splitting or cloning only if it satisfies both the standard gradient criterion and a support criterion:

$$\left(\|\nabla_\mu^{(k)}\|_2 > \tau_{\text{grad}}\right) \wedge \left(s_k > \tau_{\text{mid}}\right), \tag{7}$$

Fig. 3. Qualitative stability under trajectory deviation. We compared novel view synthesis on Mip-NeRF 360 and DL3DV-10K across a progressive increase in the viewpoint change: View 1 (top) remained near the training trajectory, whereas Views 2 and 3 (middle and bottom) deviated further with reduced input coverage. Under increasing deviation, FSGS [4], SparseGS [3] and NexusGS [7] exhibit representation-level failures, including background collapse and billboard-like stretching. In contrast, our Evidence-Gated Stabilization shows fewer visible artifacts in these examples and preserves sharper local appearance under viewpoint deviation.

where τ_{grad} follows the standard 3DGS densification threshold. This rule biases topology growth toward primitives with stronger multi-view support and discourages capacity allocation in weakly supported regions.

Support-Based Pruning. Pruning is conservative: a primitive is removed only when it has both low support and low opacity,

$$(\alpha_k < \tau_\alpha) \wedge (s_k < \tau_{low}) \,, \tag{8}$$

where α_k is opacity. Combining support with opacity avoids removing valid but weakly observed primitives solely because their angular support is low.

E. Fixed-Topology Refinement

After topology adaptation, we disable densification and run a fixed-topology refinement stage. We maintain an Exponential Moving Average (EMA) of the normalized support score, which reduces sensitivity to transient visibility or scaffold fluctuations. During refinement, opacity is annealed as

$$\alpha_k \leftarrow \alpha_k \cdot (\epsilon_\alpha + (1-\epsilon_\alpha)\bar{s}_k)^{\gamma_\alpha} \,, \tag{9}$$

where $\epsilon_\alpha = 0.1$ prevents hard deletion, and $\gamma_\alpha \geq 1$ controls attenuation strength. High-support primitives are nearly unchanged, while persistently low-support primitives are gradually attenuated.

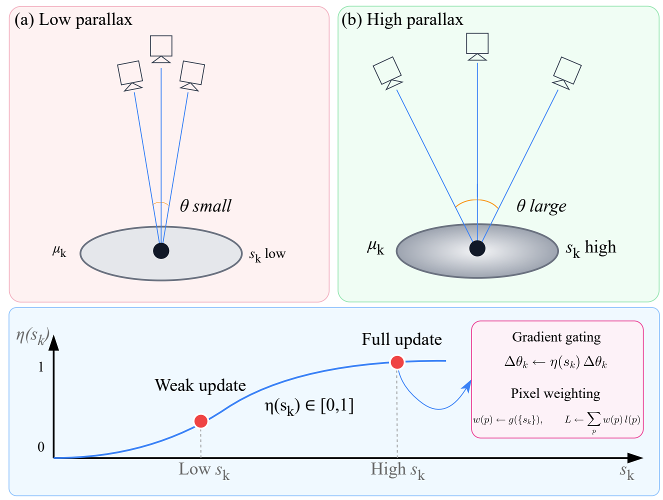

Fig. 4. Geometric evidence scoring and gradient gating. Top: The Angular Support Score s_k assigns lower support to narrow-baseline observations and higher support to primitives observed with wider parallax. Bottom: The gate $\eta(s_k)$ modulates geometry updates; low-support primitives receive suppressed photometric geometry gradients, while higher-support primitives receive stronger updates.

IV. Experiments

A. Setups

Datasets and Protocol. We evaluate dataset-specific sparse-view settings: Tanks and Temples with 4 views, DL3DV-10K with 8 views, and Mip-NeRF 360 with 12 views. For a target of N views from an ordered sequence of T frames, we select every $\lfloor T/N \rfloor$-th frame, starting from the first frame, until N views are selected. Held-out frames form the evaluation set. Within each dataset, all re-run methods use identical input views, camera parameters, initialization, and training budget. Because view count is tied to dataset, these results do not isolate the effect of view count across datasets. View-sampling robustness is evaluated in Sec. IV-D (Table IV).

Metrics. We report PSNR, SSIM, and LPIPS on held-out views with ground-truth images. To better assess semantic and structural fidelity, often missed by pixel-wise metrics, we also include an auxiliary VLM-based perceptual score from LLaVA [43], following VistaDream [44]. Additionally, we distinguish between (i) *near-trajectory* views (interpolated poses within the training baseline) and (ii) *off-trajectory* views (poses deviating $> 15°$ from the nearest training camera), which better reveal representation-level artifacts, such as floaters or structural distortions.

Baselines. We compared against vanilla **3DGS** [1] to demonstrate the instability of unregularized optimization. For state-of-the-art benchmarking, we evaluate against a comprehensive set of sparse-view variants, including depth-regularized methods like **FSGS** [4] and **DNGaussian** [2], as well as recent geometry-aware approaches such as **SparseGS** [3], **CoR-GS** [6], **CoMapGS** [5], and **NexusGS** [7]. All re-run baselines use identical input images and camera parameters while retaining their standard training protocols. CoMapGS is the exception: because its code is unavailable, its Mip-NeRF 360 result is quoted from the original paper [5] and is treated only as a reference.

Method	Tanks & Temples (4 Views)			DL3DV-10K (8 Views)			Mip-NeRF 360 (12 Views)		
	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓
3DGS [1]	14.18	0.628	0.301	16.54	0.465	0.462	17.49	0.499	0.531
FSGS [4]	18.36	0.707	0.232	18.44	0.540	0.311	18.15	0.504	0.485
DNGaussian [2]	18.91	0.790	0.176	17.13	0.481	0.492	16.28	0.512	0.509
CoR-GS [6]	18.74	0.798	0.181	19.42	0.586	0.358	18.46	0.531	0.412
CoMapGS† [5]	–	–	–	–	–	–	19.680	0.591	0.394
NexusGS [7]	19.43	0.819	0.163	20.27	0.631	0.319	19.21	0.564	0.352
SparseGS [3]	18.89	0.834	0.178	20.81	0.653	0.423	19.37	0.577	0.398
Ours	**19.85**	**0.865**	**0.155**	**22.44**	**0.721**	**0.263**	**20.17**	**0.739**	**0.273**

TABLE I

QUANTITATIVE COMPARISON IN DATASET-SPECIFIC SPARSE-VIEW SETTINGS: TANKS AND TEMPLES WITH 4 VIEWS, DL3DV-10K WITH 8 VIEWS, AND MIP-NERF 360 WITH 12 VIEWS. ALL RE-RUN METHODS USE IDENTICAL INPUT VIEWS AND CAMERA PARAMETERS WHILE RETAINING THEIR STANDARD TRAINING PROTOCOLS. BEST, SECOND-BEST, AND THIRD-BEST RESULTS ARE HIGHLIGHTED IN RED, ORANGE, AND YELLOW, RESPECTIVELY. †COMAPGS RESULTS ARE QUOTED FROM ITS ORIGINAL PAPER BECAUSE ITS CODE IS UNAVAILABLE; THEY ARE INCLUDED FOR REFERENCE AND ARE NOT PART OF THE CONTROLLED RE-RUN COMPARISON.

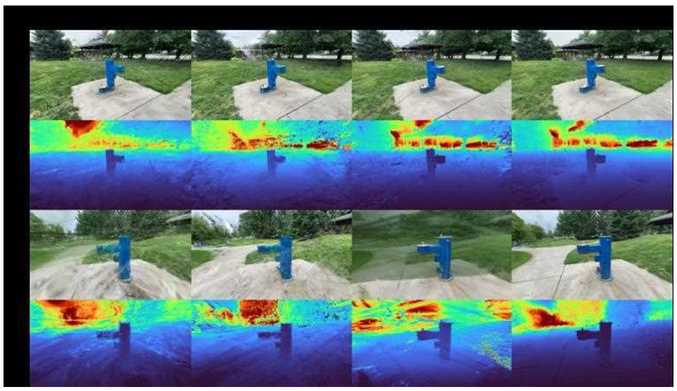

Fig. 5. Novel-view RGB renderings and accumulated depth maps on DL3DV-10K under sparse inputs. FSGS [4], SparseGS [3], and NexusGS [7] produce plausible RGB appearances, but their rendered depth maps show floating artifacts and blurred depth boundaries that intensify under viewpoint deviation. Our method yields cleaner rendered depth with sharper discontinuities and fewer visible floaters in this example, suggesting improved stability beyond appearance fitting.

Method	Noise-Free ↑	Edge ↑	Structure ↑	Detail ↑	Quality ↑
3DGS [1]	0.75	0.69	0.73	0.68	0.72
FSGS [4]	0.79	0.74	0.77	0.78	0.76
DNGaussian [2]	0.81	0.77	0.80	0.75	0.79
CoR-GS [6]	0.82	0.78	0.81	0.76	0.80
NexusGS [7]	0.84	0.81	0.83	0.79	0.82
SparseGS [3]	0.86	0.82	0.84	0.80	0.83
Ours	**0.92**	**0.89**	**0.92**	**0.88**	**0.91**

TABLE II

AUXILIARY LLAVA-IQA PERCEPTUAL EVALUATION ON OFF-TRAJECTORY VIEWS WITHOUT GROUND-TRUTH REFERENCES, EVALUATED ON MIP-NERF 360 (12 VIEWS). WE REPORT THE FRACTION OF "YES" RESPONSES FOR FIVE PERCEPTUAL CRITERIA, AVERAGED OVER ALL EVALUATED VIEWS. SEE THE SUPPLEMENTARY MATERIAL FOR THE FULL PROTOCOL AND PROMPTS.

B. Results Analysis

Quantitative Comparison. Table I shows that our method achieves the best PSNR, SSIM, and LPIPS among the controlled re-runs in each dataset-specific setting. Relative to the strongest re-run PSNR baseline, it improves by $+0.42$ dB over NexusGS on Tanks and Temples, $+1.63$ dB over SparseGS on DL3DV-10K, and $+0.80$ dB over SparseGS on Mip-NeRF 360. Its Mip-NeRF 360 PSNR is also 0.49 dB higher than the externally reported CoMapGS result. These results support the effectiveness of per-primitive evidence gating within the evaluated sparse-view settings.

These improvements in geometric consistency are further reflected in the LLaVA-IQA perceptual evaluation results shown in Table II. Our method achieves substantially higher scores in both Structure" and Quality," indicating that angular support filtering effectively suppresses the floating artifacts commonly observed in baseline methods under viewpoint changes.

Qualitative Comparison. Figures 1 and 3 show that several baselines remain plausible near the training trajectory but break down under viewpoint deviation. FSGS [4] and SparseGS [3] exhibit floaters and blurred depth discontinuities (Fig. 5). Figures 8 and 9 further indicate that CoR-GS [6], and NexusGS [7] reduce some failure cases but still show background collapse or local view-dependent blur. In contrast, Evidence-Gated Stabilization suppresses spurious primitives, producing cleaner depth maps and more stable RGB renderings. This supports the claim that angular-support-driven gating reduces view-dependent degeneracy.

Viewpoint-Deviation Degradation Analysis. To further analyze viewpoint-extrapolation behavior, Fig. 7 plots PSNR as a function of the angular deviation from the nearest training view. As the viewpoint moves away from the sparse training-camera manifold, all methods degrade, but the baselines generally exhibit faster performance drops after the $15°$ off-trajectory threshold. In contrast, our method shows a more gradual degradation curve and maintains a larger margin in the off-trajectory range. This trend is consistent with our hypothesis that regulating primitive motion and topology growth by multi-view support reduces the influence of weakly supported primitives under larger viewpoint changes.

C. Implementation Details

Our PyTorch implementation builds on the official 3DGS codebase. We train for 30,000 iterations on an RTX 4090. Key hyperparameters include anisotropy weight $\beta = 0.1$ ($\kappa = 4.0$), and dynamically updated support quantiles (20$^{\text{th}}$ and 50$^{\text{th}}$ percentiles). Full hyperparameter details, topology schedules, and depth tolerances are provided in the supplementary material.

D. Ablation Studies

Component Ablation. Table III and Fig. 6 present the additive ablation on Mip-NeRF 360. With the MASt3R scaffold

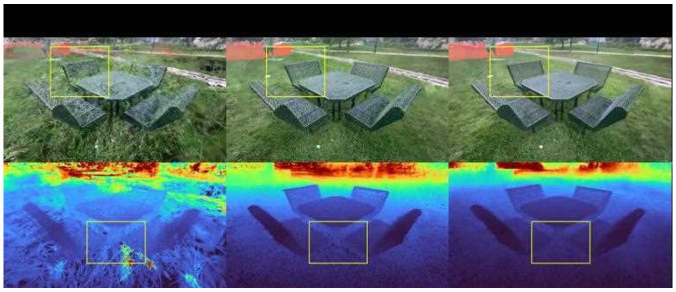

Fig. 6. Qualitative ablation on geometric stability. The baseline suffers from billboard-like artifacts and geometric distortions. The addition of Support-gated optimization sharpens RGB rendering but retains high-frequency "floaters" in the depth map. The full method, with Evidence-Guided Topology, restricts density to verified regions, suppresses floaters, and stabilizes geometry.

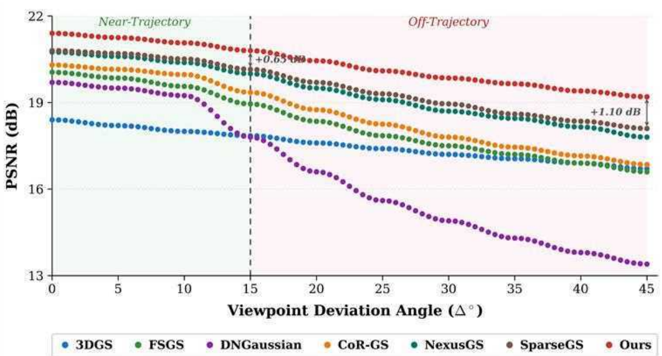

Fig. 7. **Viewpoint-deviation degradation analysis.** PSNR as a function of the angular deviation Δ° from the nearest training view. The dashed line denotes the 15° near/off-trajectory threshold. Compared with sparse-view baselines, our method degrades more gradually as viewpoint deviation increases, suggesting improved stability under off-trajectory rendering.

fixed, vanilla 3DGS obtains an LPIPS of 0.497. Adding Support-Gated Optimization reduces LPIPS to 0.360, representing the largest incremental perceptual improvement. Adding Evidence-Guided Topology further increases SSIM from 0.639 to 0.739 and reduces LPIPS to 0.273, consistent with the qualitative reduction of floaters shown in Fig. 6.

Fixed-Scaffold Attribution. Table III holds the MASt3R scaffold, input views, camera poses, and training budget fixed. Under this controlled setting, vanilla 3DGS reaches 18.11 dB PSNR, whereas the full method reaches 20.17 dB, a gain of +2.06 dB. This comparison isolates the contribution of the proposed optimization and topology policy within the tested scaffold.

Score Design and Gating Scope. Removing the resolution-confidence term and using only $\sin\theta$ reduces PSNR from 20.17 to 19.66 dB, supporting the inclusion of baseline-to-depth weighting in the support score. Enforcing strict isotropy (κ=1) instead of the fixed anisotropy bound κ=4 reduces PSNR to 19.52 dB, indicating the benefit of permitting bounded anisotropy. Extending support gating to both geometry and appearance reduces PSNR to 19.74 dB, supporting the decoupled design in Sec. III-C: only photometric geometry gradients are support-gated, while appearance and opacity updates remain ungated and geometry regularization remains

Ablation Variant	PSNR ↑	SSIM ↑	LPIPS ↓
Component Additions			
Baseline (MASt3R Scaffold + Vanilla 3DGS)	18.11	0.514	0.497
+ Support-Gated Opt.	19.33	0.639	0.360
+ Evidence-Guided Topo. (Ours)	**20.17**	**0.739**	**0.273**
Score Design & Gating Scope			
s_k w/o resolution-confidence ($\sin\theta$ only)	19.66	0.706	0.307
Strict isotropy (κ=1) instead of κ=4	19.52	0.698	0.318
Global gating (geometry + appearance)	19.74	0.713	0.298

TABLE III
ABLATION ON MIP-NERF 360 (12 VIEWS). THE FULL MODEL (GRAY ROW) UTILIZES THE MAST3R SCAFFOLD, DECOUPLED GATING, AND CONTINUOUS SCORE METRICS.

Sampling	Vanilla 3DGS		Ours	
	PSNR	Δ	PSNR	Δ
Uniform (default)	17.49	–	**20.17**	–
Random	16.84	−0.65	19.85	−0.32
Clustered (hemisphere)	14.39	−3.10	18.94	−1.23

TABLE IV
ROBUSTNESS TO VIEW-SAMPLING PATTERN ON MIP-NERF 360 (12 VIEWS). ADAPTIVE QUANTILE THRESHOLDS ABSORB MOST OF THE DISTRIBUTION VARIATION; VANILLA 3DGS DEGRADES 2.5× AS MUCH UNDER CLUSTERING.

active.

Robustness to Camera Distribution. We stress-test 12-view sampling strategies on Mip-NeRF 360 (Table IV). Under severe clustering (single hemisphere), median support drops $\sim$30%, automatically tightening the adaptive quantiles (τ_{low}, τ_{mid}). Consequently, our method degrades by only 1.23 dB, whereas vanilla 3DGS collapses by 3.1 dB (2.5× worse). On the forward-facing LLFF Fern, Flower, and Fortress scenes with identical 8-view inputs, poses, initialization, and training budget, the three-scene averages are 18.62/0.571/0.432 for vanilla 3DGS, 19.09/0.611/0.393 for FSGS, 19.29/0.627/0.373 for SparseGS, and 19.41/0.643/0.348 for ours (PSNR/SSIM/LPIPS). The smaller gain relative to wide-baseline scenes is consistent with the method's dependence on available parallax, while ours remains best across all three averaged metrics. Finally, s_k is not a self-fulfilling regularizer: on *externally trained* 3DGS and FSGS models, it strongly correlates with per-primitive held-out error (Spearman ρ=−0.68 and ρ=−0.61; see supplementary).

V. LIMITATIONS AND DISCUSSION

Although Evidence-Gated Stabilization improves sparse-view 3DGS, limitations remain. First, failures in the geometric scaffold or severe pose errors propagate to support estimation and can cause incorrect gating. Second, under near-collinear viewpoints, vanishing parallax weakens the support signal and reduces the advantage over vanilla 3DGS. Third, distant background regions may receive lower support because of the baseline-to-depth normalization, occasionally causing background blur; distance-aware normalization is a promising future direction. Regarding efficiency, angular support scores are recomputed every $T_{\text{supp}} = 500$ iterations. The support variables are used only during training and are discarded at inference.

VI. Conclusion

We presented Evidence-Gated Stabilization, a reliability-aware optimization framework for sparse-view 3DGS. Our central observation is that sparse-view failures arise not only from limited supervision, but also from applying the same geometry updates and topology growth rules to well-supported and weakly supported primitives. By estimating primitive-level multi-view support and using it to regulate geometry updates, densification, pruning, and opacity refinement, our method reduces the influence of weakly supported primitives in the evaluated sparse-view settings. The approach remains dependent on scaffold quality and available parallax, but provides a practical mechanism for incorporating multi-view reliability into sparse-view 3DGS optimization.

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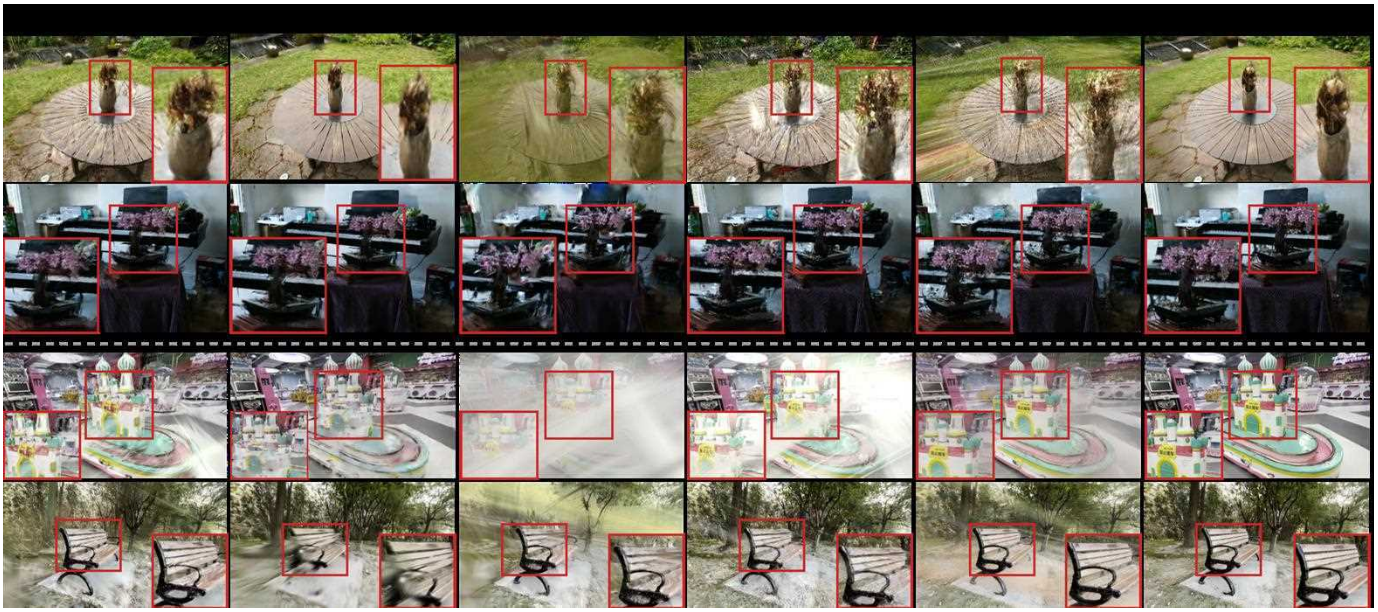

Fig. 8. Unified qualitative comparison on off-trajectory novel views across Mip-NeRF 360 (top two rows, 12 training views) and DL3DV-10K (bottom two rows, 8 training views). Perceptual quality is quantified in Table II. Baselines exhibit distinct failure modes under viewpoint deviation: FSGS [4] produces billboard-like stretching and foreground floaters caused by unconstrained gradient updates in low-parallax regions; CoR-GS [6] reduces floaters but suffers background collapse and view-dependent color bleeding; SparseGS [3] exhibits planar surface stretching as the viewpoint deviates from the training manifold; NexusGS [7] improves structural coherence via epipolar depth priors but retains residual blur on fine foreground structures.

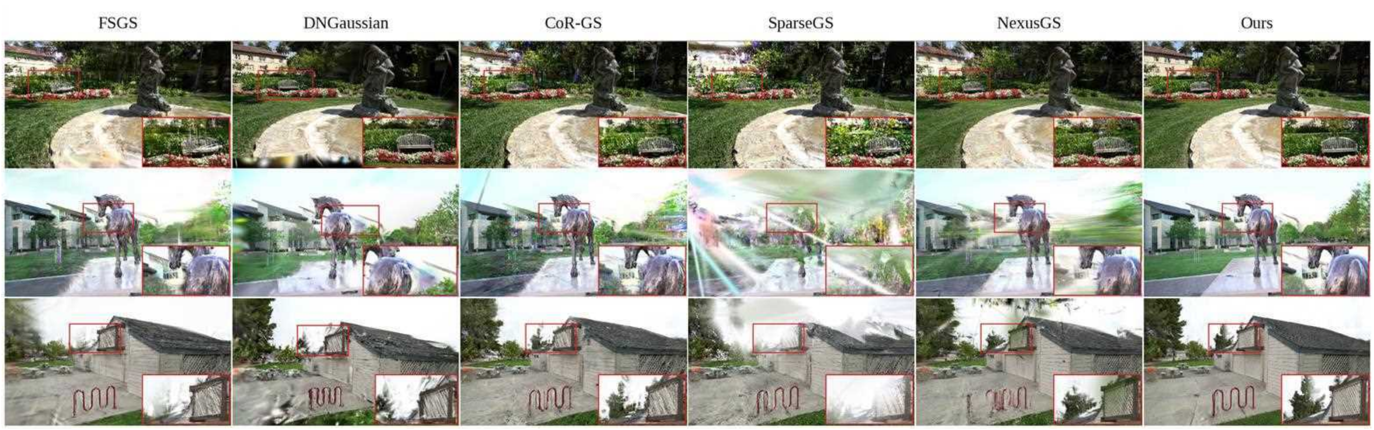

Fig. 9. Qualitative comparison on Tanks and Temples [45] (4 training views) across *Ignatius*, *Horse*, and *Barn* scenes under off-trajectory novel views. FSGS [4] exhibits foreground floaters and blurred surface boundaries; CoR-GS [6] reduces floaters but introduces view-dependent color bleeding visible in reflective and thin structures; SparseGS [3] produces planar billboard stretching under significant viewpoint deviation; NexusGS [7] improves structural coherence but retains residual blur in background regions. Our Evidence-Gated Stabilization suppresses these artifacts by restricting geometric updates and densification to multi-view-verified regions, yielding sharper structure and more coherent renderings across all scenes.